MECHTRON / SFWRENG 3K04: Software Development

Project Description – Fall 2026

McMaster University

Part A: Automated Dispensing Cabinet (ADC)

1 Project Context

This project focuses on the development of a software system for an **Automated Dispensing Cabinet (ADC)** used in hospital environments. An ADC is a safety-critical, software-controlled dispensing device.

The ADC is one part of a larger hospital medication management solution. The ADC software operates locally at the patient care location and is responsible for controlling cabinet operation, managing medication access, monitoring hardware conditions, and recording operational activities.

Moreover, an ADC is not a complete hospital medication-management application. Instead, it acts as the software brain of the cabinet and works together with a separate Monitoring System. The Monitoring System is another software system based on a human-facing application used by nurses, pharmacists, technicians, and other authorized people to view alerts, manage schedules, authorize actions, and respond to failures.

Therefore, the ADC and the Monitoring System, while two separate software systems developed independently, must work together and be integrated later. The purpose of this integration activity is to allow two independently designed software systems to communicate and exchange information as would be expected in real-life application scenarios.

Regardless of the part each sub-team is assigned, sub-teams are expected to analyze the provided information, identify missing details, make reasonable assumptions, and develop system requirements and design solutions. Both Parts A and B focus on system analysis and design covering the software development lifecycle, not only software implementation.

2 Introduction

Hospitals use medication dispensing cabinets to provide healthcare workers with secure and timely access to medications. These cabinets help improve medication management by controlling access, maintaining inventory information, and recording medication activities.

A hospital plans to introduce a software-controlled ADC. The ADC will operate inside hospital wards and will communicate with a centralized Monitoring System used by pharmacy personnel and hospital administrators.

It is important to realize that the ADC is a safety-critical device because incorrect behaviour and malfunction may lead to harm. Examples include dispensing the wrong medication, dispensing at the wrong time, dispensing multiple doses, failing to make a required dose available, allowing unauthorized access to medication, hiding a device failure, or incorrectly reporting that a dose was

dispensed or removed.

The goal of Part A is to design the embedded software system responsible for controlling the ADC operation. A secondary intended objective is to pair this ADC with the Monitoring System system through appropriate integration, which requires ADC teams and Monitoring System teams working together on the interface definitions for their proper communication.

3 Project Scope and Assumptions

As Part A focuses on a software solution responsible for controlling an ADC, it is important to understand its overall scope and assumptions.

3.1 Project Scope

The ADC software is responsible for managing:

- accress control enforcement;
- medication dispensing operations;
- inventory updates;
- local event logging;
- fault detection;
- operational modes;
- communication with external systems.

The following items are outside the project scope:

- mechanical design of the cabinet;
- detailed design of drawers, locks, and storage mechanisms;
- hospital database systems;
- patient information systems;
- internal design of the Monitoring System;
- medication selection or medical decision processes.

The focus should be on the software behaviour of the ADC and its interaction with users, hardware components, and external systems.

3.2 Project Assumptions

Sub-teams should assume that:

- the physical cabinet hardware already exists;
- sensors and actuators provide software interfaces;
- the hospital network is available;
- a means of capturing user credentials at the cabinet is available;
- medications are loaded by authorized personnel.

Any additional assumptions and decisions should be documented.

4 System Overview

The ADC is an embedded software system that controls a secure medication storage cabinet.

The software communicates with hardware components such as:

- electronic locks;
- drawers;
- sensors;
- actuators;
- local user interfaces.

The ADC software manages medication access, monitors cabinet conditions, updates inventory information, and records operational events.

The system should provide safe operation even when communication with external systems is temporarily unavailable.

The ADC communicates with a centralized Monitoring System. The detailed design of the Monitoring System is intentionally outside the scope of this project.

A simple block diagram showing the system overview is depicted in Figure 1.

5 Operational Environment

The ADC operates inside a hospital ward near patient care areas.

Typical users include:

- nurses;
- pharmacists;

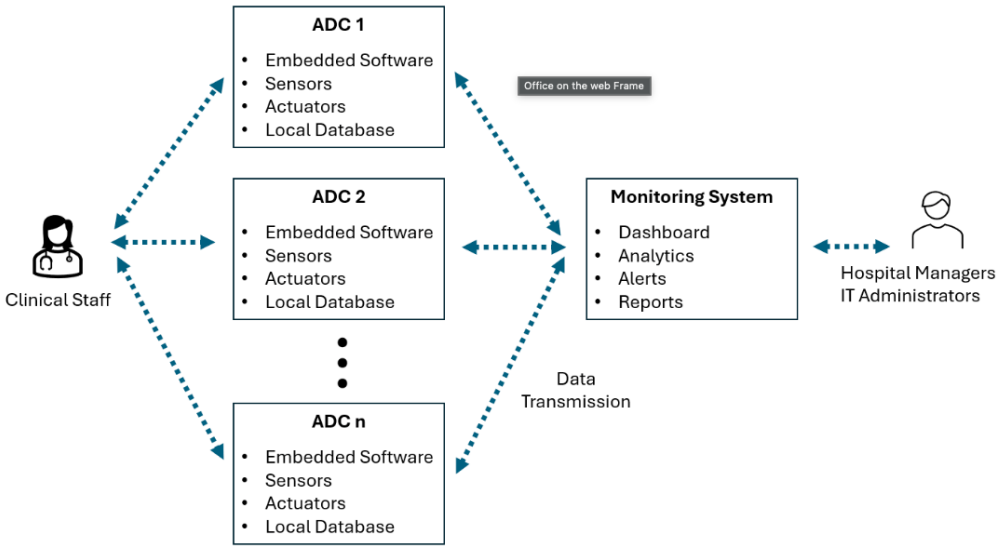


Figure 1: Simple block diagram showing the system overview.

- technician.

Different users may have different responsibilities and access permissions. The ADC must operate in an environment where:

- medication security is important;
- access must be controlled;
- activities must be traceable;
- failures must be detected.

6 High-Level System Capabilities

The ADC is expected to support the following capabilities, intentionally described in high level terms. Your team must use them as reference points for the development of a software solution.

- **Access Control:** The ADC should ensure that only permitted users can perform protected actions.
- **Medication Inventory Management:** The ADC should maintain information about medications stored inside the cabinet.
- **Medication Dispensing:** The ADC should support controlled access to medications based on authorized requests.
- **Event Logging:** The ADC should record important operational activities.
- **Fault Detection:** The ADC should detect abnormal hardware and software conditions.
- **Operational Mode Management:** The ADC should support different operating conditions such as normal operation, maintenance, and fault conditions.

- **External Communication:** The ADC should exchange operational information with the Monitoring System.

7 Example Operational Scenarios

7.1 Scenario 1: Medication Dispensing

A nurse authenticates at the ADC and requests medication for a patient. The ADC verifies authorization, provides access to the required medication location, updates inventory information, and records the activity.

7.2 Scenario 2: Inventory Replenishment

A pharmacy technician accesses the ADC and adds new medication inventory. The system updates internal inventory information and records the activity.

7.3 Scenario 3: Hardware Fault

A drawer does not close correctly after use. The ADC detects the problem and determines the appropriate system response.

7.4 Scenario 4: Communication Loss

The ADC loses communication with the Monitoring System. The ADC continues local operation and manages information that cannot be immediately transmitted.

8 System Constraints

The ADC design should consider the following constraints:

- Only authorized users can access medications.
- Inventory information must remain accurate.
- Important activities must be recorded.
- The system must operate safely during communication failures.
- Hardware and software faults must be detected.
- The system must operate reliably in a hospital environment.

9 External Interfaces

The ADC communicates with the Monitoring System. Information exchanged may include:

- operational status;
- inventory information;
- alarms;

- maintenance information;
- system health information.

The interface requirements between ADC and Monitoring System are intentionally not provided. A few integration considerations are provided below to help teams think about this problem.

9.1 Integration Considerations

The ADC team will work with a Monitoring System team to define the communication interface between both systems.

The ADC team is responsible for defining:

- information that the ADC provides;
- information that the ADC requires;
- communication requirements;
- local handling of communication failures;
- synchronization requirements.

The final goal is to demonstrate that the ADC software can communicate correctly with the Monitoring System.